

SECURITY ANALYSIS UW ORBITAL

Attack 1- Attacker gains control of the satellite

More details in *Figure 1*.

The ultimate goal of the attack is to gain control of the satellite which means executing commands, resetting the satellite, deleting data...

In the major branches, it can be found:

1. Send unauthorized command to satellite
2. Modify satellite firmware

About the **first major branch**, this means the attacker does not change firmware, but instead sends commands that the satellite will execute (CMD_MICRO_SD_FORMAT). Following the branch, we find:

1.1. Gain ability to transmit command packets means that before sending commands, the attacker must first get access to the communication channel. To achieve this, the attacker can 1.1.1. Access ground station system or 1.1.2. Build their own transmitter.

To compromise the GS system 1.1.1, the attacker can 1.1.1.1. Steal operator credentials (phishing, password reuse, leaked credentials) or 1.1.1.2. Exploit backend vulnerability (using FASTAPI: possible injection attacks if SQL, authentication bypass, insecure API endpoints) or 1.1.1.3. Use a compromised developer machine (developer laptop infected, attacker gets access to repository and command tools).

if an attacker builds an independent radio transmitter 1.1.2, the attacker can bypass GS. The steps for this are 1.1.2.1 Reverse engineer command protocol (commands ID, packet structure, packing function - packUInt32() packets binary encoded), and 1.1.2.2. Construct valid command packets (can be done once the protocol is known), and 1.1.2.3. Transmit packet to satellite (with radio hardware, using the correct frequency and antenna).

Another branch 1.2. Bypass command authentication (satellite should verify packets). If 1.2.1. No authentication implemented, satellite executes any packet it receives. Also the attacker can 1.2.2. Replay previously captured command (replay attack - CMD_DOWNLINK_TELEM) or 1.2.3. Forge command packet fields by modifying commands id, parameters or timestamp.

About the **second major branch**, this is more powerful but harder, instead of sending commands, the software is replaced. This can be done 2.1. Upload malicious firmware, or 2.2 Exploit firmware updates (if satellite allows remote updates).

2.1. can be performed by 2.1.1. Access firmware flashing interface (UniFlash), or 2.1.2. Modify firmware binary, inserting malicious code (hiding commands, backdoor or

disabling authentication in case it is), or 2.1.3. Flash firmware through bootloader (if update is not signed, attacker can install anything).

2.2. can be performed by 2.2.1. Send malicious update command (CMD_SET_PROGRAMMING_SESSION), 2.2.2. Bypass update verification (if firmware is not cryptographically signed, attacker update may succeed), or 2.2.3. Trigger remote firmware update (making the satellite install malicious firmware).

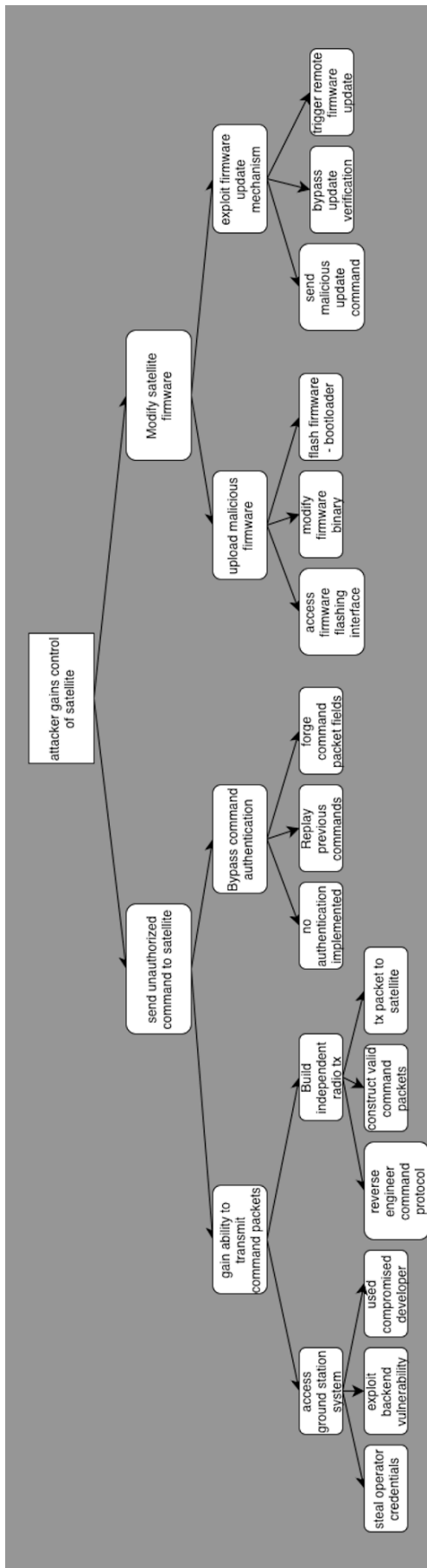


Figure 1: Attack 1- Attacker gains control of the satellite